

The KNN neighbour-count sweep

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Problem 1. *With plain cosine similarity, weights $w_j = \max(s_j, 0)^\alpha$ ($\alpha = 1$) and risk score $R_k = \sum_{j \leq k} w_j y_j / \sum_{j \leq k} w_j$, the full-pool weights are nearly uniform (similarities 0.83–0.95; effective sample size 34,032 of 34,063). Why does the test-set ROC AUC at the whole pool end at 0.608 rather than falling to 0.5?*

Solution. The weights of the nearer neighbours, who should have more influence, are still larger. That is not the case with exactly uniform weighting. The effect is stronger with importance-weighted cosine similarity.